

Overtune 3 (ot3) — System Installation Guide

NATIVE DEPLOYMENT, IN-APP AUTO-UPDATER, AND PORTABLE SHAREWARE BUNDLES

1. System Overview & Hardware Requirements

Overtune 3 (ot3) is a high-precision digital filter design and live audio simulation workstation built on pure C++20 and Qt 6. It features automated Stage 2 compliance mathematical audits, sub-millidecibel magnitude precision, and SIMD-accelerated cascaded biquad evaluation.

- **Linux Requirements:** Any modern 64-bit Linux distribution (Ubuntu 22.04+, Fedora 38+, Arch Linux, Debian 12+) with GLIBC 2.34+ and Qt 6.4+ libraries.
- **Windows Requirements:** Windows 10 (64-bit, 1809+) or Windows 11 (x64) with Microsoft Visual C++ 2022 Redistributable.
- **Display:** Minimum recommended display resolution of 1280 × 800 with HiDPI display scaling support.

2. Linux Installation & Execution

Overtune 3 provides three straightforward deployment methods on Linux platforms:

Method A: Native Automated Setup (install_linux.sh)

The interactive installer integrates **ot3** into your desktop environment, creates desktop and application menu entries, registers icons, and links the CLI binary:

```
# User-level installation (No root required - installs to ~/.local/share/
overtune3)
./scripts/install_linux.sh

# System-wide installation (Installs to /opt/overtune3 and /usr/local/bin)
sudo ./scripts/install_linux.sh
```

Once completed, launch Overtune 3 from your application menu or directly run **ot3** from any terminal.

Method B: Standalone Portable Shareware Bundle

For zero-install portable deployment (e.g., from a USB drive or local directory):

```
tar -xzf overtune3-linux-x86_64.tar.gz
cd overtune3-linux-x86_64
./ot3.sh
```

Method C: Building from Source

```
./build.sh          # Builds optimized release binary in ./build/bin  
    /ot3  
./build/bin/ot3     # Direct launch
```

3. Windows Installation & Execution

Overtune 3 provides native PowerShell and NSIS installer packaging for Windows:

Method A: PowerShell Automated Setup (install_windows.ps1)

Runs natively on any standard Windows 10/11 system without requiring external package managers:

```
powershell -ExecutionPolicy Bypass -File .\scripts\install_windows.ps1
```

This installs `ot3.exe` to `%LOCALAPPDATA%\Programs\Overtune3`, creates Start Menu and Desktop shortcuts, registers the uninstaller in Windows Settings / Control Panel, and configures the background updater helper.

Method B: Modern NSIS Setup Executable

Run `Overtune3-Setup-x64.exe` and follow the graphical setup wizard. Silent unattended deployment is supported with the `/S` switch.

Method C: Standalone Portable ZIP

Extract `overtune3-windows-x64.zip` and double-click `bin\ot3.exe`.

4. In-App Auto-Updater & In-Place Atomic Updates

Overtune 3 features a unified OLED modal updater window matching the What's New and Stage 2 Verifier interfaces. It handles live patch updates seamlessly:

- **Accessing Updates:** Select **Help** → **Check for Updates...** or click the **Installer / Update Ready** badge in the titlebar.
- **Linux Atomic Patching:** Leverages POSIX atomic `unlink/rename` semantics. The running application binary is replaced smoothly in place without requiring process termination before write.
- **Windows Detached Updater:** Bypasses Windows executable file-locking by spawning a detached helper script (`overtune.update.bat`) that waits for the parent process PID to exit cleanly, replaces `ot3.exe` with `ot3.exe.new`, and restarts the updated studio application.
- **Cryptographic SHA-256 Audit:** All downloaded payload chunks are validated against release cryptographic checksums before application.

5. Command-Line Options & Essential Shortcuts

Option / Key	Type	Functionality
ot3 --install	CLI Flag	Launch directly into the native System Installation manager.
ot3 --update	CLI Flag	Query release server and display available version patches.
Ctrl + 1 .. 6	Keyboard	Navigate directly between Design, Analysis, Simulation, Export, Docs, Settings.
Ctrl + 0	Keyboard	Auto-scale and reset frequency, pole-zero, and impulse response plots.
F5 / Space	Keyboard	Start or pause the live real-time audio and DSP simulation engine.
Ctrl + T	Keyboard	Toggle between OLED Dark and Studio Light visual themes.

For support, bug reports, and full DSP mathematical derivations, refer to the in-app documentation page or open the `theory_and_math.pdf` manual.